PROJECT REPORT

DESIGN AND DEVELOPMENT OF A SMART CALCULATOR IN C

Arithmetic operations using user-defined functions and control structures

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1. Abstract

This project presents the design and development of a menu-driven Smart Calculator using the C programming language. The program performs four basic arithmetic operations: addition, subtraction, multiplication, and division. Each operation is implemented through a separate user-defined function. A switch-case statement selects the operation, while a do-while loop allows the user to perform repeated calculations until the Exit option is selected. An if-else condition prevents division by zero. The project demonstrates functions, input/output operations, decision-making statements, looping, arithmetic operators, and basic error handling.

2. Introduction

A calculator is a common application used to perform mathematical calculations. Developing a calculator in C is a useful programming exercise because it combines important fundamentals in one program. The user selects an operation from a menu and enters two numbers. The program calls the corresponding function, displays the result, and then shows the menu again.

The program is text-based and designed to be straightforward, readable, and easy to modify. Dividing the arithmetic operations into separate functions makes the code organised and reusable.

3. Objectives

- To develop a menu-driven calculator using C.
- To perform addition, subtraction, multiplication, and division.
- To implement each arithmetic operation using a separate user-defined function.
- To apply switch-case and if-else control structures.
- To use a do-while loop for repeated calculations.
- To handle division by zero and invalid menu choices.

4. Software and Hardware Requirements

Category	Requirement
Programming language	C
Compiler	GCC, Clang, or another standard C compiler
Editor / IDE	Visual Studio Code, Code::Blocks, Dev-C++, or any text editor
Hardware	A computer or laptop
Operating system	Windows, macOS, or Linux

5. Concepts Used

5.1 User-defined functions: A function is a reusable block of code that performs a specific task. This program uses four functions: add(), subtract(), multiply(), and divide(). Each function accepts two double values and returns the calculated result.

5.2 Function declaration and definition: Function declarations are written before main() so the compiler knows each function's return type and parameters. Function definitions are written after main() and contain the calculation logic.

5.3 switch-case statement: The switch statement checks the user's menu choice and executes the matching case. The break statement prevents execution from continuing into another case.

5.4 do-while loop: The do-while loop displays the menu at least once and repeats it until the user enters 5 to exit.

5.5 if-else statement: Before division, the program checks whether the second number is zero. If it is zero, an error message is displayed instead of performing the division.

5.6 Input and output: printf() displays instructions and results, while scanf() accepts the user's menu choice and numbers.

6. Algorithm

1. Start the program.
2. Declare variables for the menu choice, two input numbers, and result.
3. Display the calculator menu.
4. Read the user's choice.
5. If the choice is from 1 to 4, accept two numbers.
6. Use switch-case to identify the selected operation.
7. Call the corresponding user-defined function.
8. For division, check whether the second number is zero before calling divide().
9. Display the result or an error message.
10. Repeat the menu until the user selects Exit (5).
11. Stop the program.

7. Flow of the Program

Step	Action
1	Start
2	Display menu and read choice
3	Read two numbers for choices 1 to 4
4	switch-case selects the operation
5	Call the matching user-defined function
6	Check divisor is not zero for division
7	Display result or error message
8	Repeat until Exit is selected
9	Stop

Textual flow: Start → Display menu → Read choice → Input numbers → Select operation → Call function → Validate division → Display output → Repeat or Exit.

8. Source Code

The following is the complete source code used in this project. It implements the four required arithmetic operations using separate user-defined functions and control structures.

```
#include <stdio.h>

// Function Declarations
double add(double a, double b);
double subtract(double a, double b);
double multiply(double a, double b);
double divide(double a, double b);

int main()
{
    int choice;
    double num1, num2, result;

    do
    {
        printf("\n====================================\n");
        printf("          SMART CALCULATOR\n");
        printf("====================================\n");

        printf("1. Addition (+)\n");
        printf("2. Subtraction (-)\n");
        printf("3. Multiplication (*)\n");
        printf("4. Division (/)\n");
        printf("5. Exit\n");

        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        if (choice >= 1 && choice <= 4)
        {
            printf("Enter first number: ");
            scanf("%lf", &num1);

            printf("Enter second number: ");
            scanf("%lf", &num2);
        }

        switch (choice)
        {
            case 1:
                result = add(num1, num2);
                printf("\nResult = %.2lf\n", result);
                break;

            case 2:
                result = subtract(num1, num2);
                printf("\nResult = %.2lf\n", result);
                break;

            case 3:
                result = multiply(num1, num2);
                printf("\nResult = %.2lf\n", result);
                break;

            case 4:
                if (num2 != 0)
                {
                    result = divide(num1, num2);
                    printf("\nResult = %.2lf\n", result);
                }
                else
                {
                    printf("\nError! Division by zero is not allowed.\n");
                }
                break;

            case 5:
                printf("\nThank you for using Smart Calculator!\n");
                break;

            default:
                printf("\nInvalid choice! Please try again.\n");
        }

    } while (choice != 5);

    return 0;
}

// Function Definitions
double add(double a, double b)
{
    return a + b;
}

double subtract(double a, double b)
{
    return a - b;
```

```
}

double multiply(double a, double b)
{
    return a * b;
}

double divide(double a, double b)
{
    return a / b;
}
```

9. Explanation of the Source Code

Header file: #include <stdio.h> provides standard input and output functions such as printf() and scanf().

Function declarations: The declarations of add(), subtract(), multiply(), and divide() inform the compiler about the functions before they are called in main().

main() function: Program execution begins from main(). It declares the variables and controls the calculator menu.

Input statements: scanf() reads the user's choice and the two numbers. The if condition ensures that numbers are requested only for operation choices 1 to 4.

switch-case: Each case calls the appropriate function, displays the returned value, and uses break to end that case.

Division validation: The if-else condition checks num2 != 0. The divide() function is called only when the divisor is non-zero.

do-while loop: The loop repeats the menu until the user enters 5. This supports multiple calculations in a single run.

Function definitions: Each function receives two double values and returns the result of its arithmetic operation.

10. Sample Outputs

Example 1 – Addition

```
Enter your choice: 1\nEnter first number: 20\nEnter second number: 10\nResult = 30.00
```

Example 2 – Subtraction

```
Enter your choice: 2\nEnter first number: 20\nEnter second number: 10\nResult = 10.00
```

Example 3 – Multiplication

```
Enter your choice: 3\nEnter first number: 20\nEnter second number: 10\nResult = 200.00
```

Example 4 – Division

```
Enter your choice: 4\nEnter first number: 20\nEnter second number: 10\nResult = 2.00
```

Example 5 – Division by zero

```
Enter your choice: 4\nEnter first number: 20\nEnter second number: 0\nError! Division by zero is not allowed.
```

11. Test Cases

Test	Input (a, b)	Choice	Expected output
Addition	12, 8	1	20.00
Subtraction	12, 8	2	4.00
Multiplication	12, 8	3	96.00
Division	12, 4	4	3.00
Division by zero	12, 0	4	Error message
Invalid choice	—	9	Invalid choice message
Exit	—	5	Thank-you message

12. Advantages

• Simple and easy-to-understand program.
• Each operation is implemented in a separate reusable function.
• The switch-case statement makes operation selection clear.
• The do-while loop allows repeated calculations.
• The division-by-zero check prevents an invalid arithmetic operation.
• The program is easy to compile and test.

13. Limitations

• The program uses a command-line interface and does not include a graphical user interface.
• It supports four basic arithmetic operations only.
• It does not include advanced operations such as percentage, powers, or square roots.
• It does not fully validate non-numeric input entered by the user.

14. Possible Future Enhancements

The calculator can be improved by adding modulus, percentage, power, square root, and scientific operations. Input validation can be strengthened to handle letters and other invalid entries. A graphical user interface could also be developed as an extension.

15. Conclusion

The Smart Calculator project successfully demonstrates how to develop a menu-driven calculator in C using user-defined functions and control structures. It performs addition, subtraction, multiplication, and division through four separate functions. The switch-case statement manages the user's selection, the do-while loop allows repeated calculations, and the if-else condition handles division by zero. Overall, this project strengthens understanding of functions, input/output operations, decision-making, looping, and structured programming in C.

16. References

• E. Balagurusamy, *Programming in ANSI C*, McGraw Hill.
• Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*, 2nd Edition, Prentice Hall.
• Programiz, C Programming Tutorials: https://www.programiz.com/c-programming
• cppreference, C Language Reference: https://en.cppreference.com/w/c

— End of Report —

Student details and source code in this report have been updated to match the final calculator program prepared according to the stated college requirement.